# From Retained Demand to Required Liquidity Density: A Conditional Sizing Framework for Institutionally Connected Digital Assets

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## Abstract

This paper examines how retained demand, if it becomes operationally meaningful, translates into required liquidity conditions for institutionally connected digital assets. Building on Paper 1's retained-demand framework and Paper 2's settlement stack and closure-competition framework, it argues that sizing should be analytically separated from price narratives. The relevant question is not what an asset should be worth, but what level of liquidity density, inventory depth, and slippage tolerance would be required if the asset were relied upon in inventory, collateral, liquidity-buffer, or waiting-liquidity form.

The paper develops a conditional sizing framework centered on gross-flow density, usable float, available depth, turnover, and bounded market impact. It argues that sizing outputs should be treated as liquidity-feasibility tests rather than valuation estimates or target prices. It further specifies the conditions under which stronger sizing claims weaken or fail, including low retained-demand intensity, weak realized selection, netting and compression effects, internalized settlement, favorable turnover, distributed need, and full-coverage substitutes.

The contribution of the paper is therefore not to advance a directional valuation thesis. It is to clarify how retained demand may be translated into required liquidity conditions under explicit and weakenable assumptions. Although an XRP-adjacent context remains the primary motivating case in the background, the paper's contribution remains methodological rather than asset-specific.


## 1. Introduction

Digital asset analysis often reaches a familiar impasse once structural relevance is taken seriously. It may become possible to argue that an asset is not merely visible in transactional flow, that infrastructure expansion does not by itself settle the question of asset demand, and that competing settlement stacks do not necessarily eliminate every boundary-crossing friction. But even if those points are granted, a further question remains. If retained demand becomes operationally meaningful, what level of liquidity would actually be required for that relevance to function at scale?

This paper addresses that question. It examines how retained demand, if it becomes economically and operationally significant, translates into required liquidity conditions for institutionally connected digital assets. The central issue is not what such an asset should be worth in market price terms, nor whether it can attract directional valuation narratives. The issue is what level of liquidity density, inventory depth,

and tolerated slippage would be required if the asset were to support operationally meaningful functions in inventory, collateral, liquidity-buffer, or waiting-liquidity form.

Paper 1 argued that institutionally connected digital assets should not be evaluated through visible usage alone, but through the stronger question of whether institutional environments create reasons to retain them in held form. It decomposed retained demand into inventory demand, collateral demand, liquidity-buffer demand, and waiting liquidity, and separated domain expansion, asset selection, and asset indispensability as analytically distinct stages. Paper 2 extended that framework by examining whether competing settlement stacks actually close the relevant boundary-crossing problem, and by separating expansion from closure and overflow from selection. Together, those two papers identify the structural conditions under which a stronger relevance claim may remain possible. They do not, however, answer the next quantitative question: what liquidity conditions would such relevance require?

That question cannot be answered simply by referring to market capitalization, spot price, or generalized claims about future valuation. Those categories may matter for some purposes, but they do not by themselves identify whether an asset can support operationally meaningful settlement, buffering, collateralization, or staged balance requirements at non-trivial scale. A quantity problem still remains. If an asset is to be held in advance, mobilized under stress, staged across time-sensitive workflows, or relied upon under bounded execution conditions, then one must ask not only whether the asset matters in principle, but what usable depth and liquidity conditions would be required in practice.

This is why sizing must be separated from price narratives. The purpose of this paper is not to estimate target prices in disguised form. Nor is it to claim that any one asset necessarily requires a particular valuation outcome. Its narrower contribution is to provide a conditional sizing framework for institutionally connected digital assets. The framework asks how retained demand may be translated into required liquidity conditions under explicit assumptions about gross flow, usable float, turnover, inventory positioning, slippage tolerance, and bounded market impact. In that sense, the paper is concerned not with directional price advocacy, but with the liquidity conditions implied by operational relevance if that relevance is real.

The central concept introduced here is required liquidity density. Broadly stated, this refers to the liquidity conditions needed to process a given level of gross flow within bounded slippage, timing, and operational constraints. In this paper, liquidity density does not mean nominal market size. It refers to the practically usable combination of positioned balances, executable depth, turnover capacity, and acceptable market impact available at the points where institutional workflows require liquidity. This concept matters because the same nominal flow requirement may imply very different balance-sheet and execution conditions depending on available float, market depth, inventory concentration, turnover, and tolerated execution stress. A system may therefore appear large in headline terms while remaining thin in usable settlement depth, or conversely may support substantial operational flow without corresponding price narratives. The relevant audit problem is to distinguish those cases rather than collapse them into a single valuation story.

This framing also clarifies the relation between retained demand and sizing. Inventory demand implies required positioned balances. Collateral demand implies required recognized and mobilizable balances. Liquidity-buffer demand implies required reserve depth under uncertainty and discontinuity. Waiting liquidity implies required staged balances across time. Once retained demand is treated as operationally

meaningful rather than merely conceptual, it becomes a quantity problem. Paper 3 therefore does not replace the retained-demand framework of Paper 1 or the closure-competition framework of Paper 2. It extends them into the question of what liquidity conditions would be required if those earlier conditions were actually met.

The argument of this paper is deliberately conditional. It does not assume that retained demand is already strong, that external connective assets necessarily survive closure competition, or that any sizing output should be read as a target price. Instead, it asks what follows if certain structural claims are granted under explicit assumptions. It also specifies the conditions under which stronger sizing claims weaken or fail, including low retained-demand intensity, weak realized selection, internalized settlement, netting and compression effects, and full-coverage substitutes that reduce the need for external liquidity positioning. In that sense, the paper is intended as a conditional audit of required liquidity conditions, not as a valuation thesis.

Although an XRP-adjacent context remains the primary motivating case in the background, the contribution of this paper is not asset-specific advocacy. XRP appears here because it sits close to a setting in which claims about settlement significance, bridge-like relevance, and liquidity needs are especially vulnerable to overreading. That makes it a demanding interpretive environment rather than a convenient rhetorical shortcut. The framework developed here is meant to discipline that environment, not to settle it by assertion.

The contribution of this paper is deliberately layer-specific. Paper 1 established the retained-demand audit category. Paper 2 analyzed the closure and selection conditions under which residual relevance can survive. This paper adds a third layer: it asks what quantity conditions retained demand would require if it became operationally significant. It does not settle applied questions about specific asset stacks, compression design, or institutional implementation; those belong to a later inquiry. Within that boundary, the contribution is fourfold.

This paper should therefore be read as the third layer of the retained-demand audit series. Paper 1 defines the relevant demand categories, Paper 2 tests whether residual relevance can survive closure competition, and the present paper asks what liquidity-density burden would follow if those prior conditions were satisfied. The paper does not convert that burden into a price target; it treats it as an assumption-sensitive liquidity-feasibility test.

First, it separates sizing from price narratives and argues that required liquidity conditions should be treated as a distinct analytical problem. Second, it connects Paper 1's retained-demand categories to quantity requirements in inventory, collateral, liquidity-buffer, and waiting-liquidity form. Third, it introduces required liquidity density as an audit concept linking gross flow, usable float, available depth, turnover, and tolerated slippage under bounded market impact. Fourth, it embeds these claims within a weakenable and falsifiable structure that explicitly includes compression, substitution, low retained-demand intensity, and non-concentrated settlement need as conditions under which stronger sizing claims may fail.

The remainder of the paper proceeds as follows. Section 2 explains why sizing must be separated from price narratives. Section 3 connects retained demand to required liquidity conditions. Section 4 develops gross-flow density and required liquidity density as audit concepts. Section 5 presents a conditional sizing

framework under slippage constraints and bounded market impact. Section 6 specifies weakening and invalidation conditions. Section 7 clarifies the XRP-adjacent interpretation boundary. Section 8 concludes and positions the paper as a bridge to a later applied inquiry into compression and stack integration.

## 2. Why Sizing Must Be Separated from Price Narratives

A recurring problem in digital asset analysis is the tendency to collapse structural relevance into price expectation. Once an analyst begins to ask whether an asset might matter for settlement, collateral mobility, liquidity provision, or inter-stack transfer, the discussion often shifts too quickly toward valuation language. Questions about operational necessity become questions about what price an asset "should" reach, what market capitalization it would "need," or how large a speculative rerating might follow from wider adoption. But these are not the same questions, and treating them as interchangeable introduces immediate analytical distortion.

The first reason for separation is that market price is not the same thing as usable settlement capacity. A high spot price does not by itself establish that large flows can be executed within acceptable slippage. Nor does a large nominal capitalization demonstrate that sufficient float, turnover, depth, or balance-sheet positioning exists for operationally meaningful use. Conversely, a system may support meaningful functional flow under some conditions without yielding any simple implication about what price should prevail in open markets. Price may be relevant to liquidity conditions, but it is not identical to them.

The second reason is that settlement problems are quantity problems before they are valuation narratives. If an asset is to be used in inventory form, collateral form, liquidity-buffer form, or waiting-liquidity form, then the immediate issue is not how the market rhetorically values the asset, but whether enough usable balances and enough executable depth exist where and when they are needed. This requires attention to available float, turnover, positioning, timing, concentration, slippage tolerance, and market impact. Those conditions cannot be inferred reliably from broad price narratives alone.

The third reason is that price discourse tends to obscure the role of assumptions. A valuation-style claim often appears as a singular number or directional judgment. A sizing framework, by contrast, should remain conditional and assumption-sensitive. Required liquidity conditions depend on the level of gross flow being considered, the acceptable degree of execution friction, the concentration or distribution of operational demand, the extent of internalization or netting, the fraction of float that is actually usable, and the degree to which balances must be pre-positioned or buffered. Once these conditions are made explicit, the problem no longer takes the form of a simple target-price question. It becomes a framework of scenario-bound requirements.

This distinction is especially important in institutionally connected settings. Institutions do not operate only on broad valuation narratives. They operate through workflow constraints, funding limits, risk tolerances, execution rules, collateral standards, and balance-sheet disciplines. If a digital asset is to matter in such environments, it must do so under these operational conditions. The relevant question is therefore not what headline valuation the market may eventually attach to the asset, but what liquidity conditions would be required if institutions were to rely on it in held form at non-trivial scale.

For that reason, this paper treats sizing as analytically prior to valuation rhetoric. It does not ask what price would be justified in the abstract. It asks what level of liquidity density, inventory depth, and

tolerated slippage would be required under explicit assumptions about operational use. This may, in principle, have implications for how one later interprets valuation. But such implications are downstream, indirect, and conditional. They do not alter the immediate task, which is to identify required liquidity conditions rather than to forecast price.

A related problem is the common tendency to use market capitalization as a proxy for functional capacity. That shortcut is often tempting because capitalization is easy to observe and easy to compare. But a capitalization figure says little by itself about available depth at the relevant execution points, about how much float is operationally usable rather than dormant or strategically held, about how rapidly positions can be mobilized without destabilizing price, or about how inventory must be distributed across jurisdictions, venues, counterparties, and time windows. A capitalization narrative may therefore exaggerate or understate actual operational capacity. What matters for the present framework is not nominal valuation as such, but the interaction between flow requirements and usable liquidity conditions.

Market capitalization is treated here not as a proxy for valuation, but at most as a broad reference point for the upper boundary of inventory absorption. The focus remains on whether available float can support the required density without violating bounded market-impact constraints, regardless of the rhetorical price narrative attached to that float.

This is why the present paper insists on a conditional sizing discipline. Outputs derived later in this paper are not target prices in disguised form. They are conditional simulation outputs under explicit assumptions about gross flow, usable float, turnover, inventory concentration, and tolerated slippage. Their purpose is to clarify what would be required if retained demand became operationally significant, not to assert that any particular market outcome must follow. In that sense, the framework is deliberately restrictive. It seeks to prevent the quantity problem from being absorbed into rhetoric about future appreciation.

Separating sizing from price narratives also improves falsifiability. A price thesis can often survive by shifting rhetoric, revising target ranges, or appealing to diffuse future adoption. A sizing framework is more disciplined. Its claims weaken or fail where required balances are lower than assumed, where netting and internalization compress the relevant liquidity burden, where substitutes absorb the relevant function, where tolerated slippage is wider than expected, or where realized operational demand remains too shallow to generate non-trivial inventory need. The framework is therefore not merely non-directional in tone. It is structurally designed to allow weaker outcomes.

The analytical implication is straightforward. If retained demand is to be translated into stronger relevance claims, that translation should first pass through required liquidity conditions rather than through valuation rhetoric. The relevant bridge is not from adoption stories to target prices, but from operational significance to usable depth, from usable depth to conditional sizing, and only then -- if at all -- to downstream interpretation. This paper therefore proceeds by bracketing price narratives and focusing instead on required liquidity density as the more disciplined object of analysis.

## 3. From Retained Demand to Required Liquidity Conditions

If the closure competition described in Paper 2 leaves meaningful residual overflow, then the sizing problem becomes unavoidable. The quantity of required liquidity is conditioned by the boundary-crossing

burden that settlement stacks cannot internalize. Paper 1 argued that the economically relevant question for institutionally connected digital assets is not simply whether they appear in visible flow, but whether institutional environments create reasons to retain them in held form. That argument clarified why retained demand must be distinguished from usage. But once retained demand is treated as operationally meaningful rather than merely conceptual, a further implication follows. Any held-form requirement is also a quantity requirement. If an asset must be present in advance, maintained under uncertainty, posted against obligations, or staged across time-sensitive processes, then one must ask how much of it must be available, where it must be available, and under what execution conditions it can be mobilized without causing prohibitive friction.

This is the point at which retained demand becomes a sizing problem. The relevant issue is no longer only whether there exists a structural reason to hold the asset, but what liquidity conditions are required for that held-form role to function in practice. An asset may be structurally relevant in principle yet remain operationally weak if the balances required to support that role cannot be positioned, mobilized, or replenished under acceptable cost and slippage conditions. For that reason, retained demand should not be treated as a binary category. Once it becomes meaningful, it implies required balances, required usable depth, and required tolerance for execution stress.

The quantity problem is not uniform across all forms of retained demand. Different retained-demand channels imply different forms of required liquidity condition. Inventory demand implies the need for positioned balances that can support recurring operational use without disruption. The question here is not merely whether the asset can eventually be obtained, but whether it can be maintained where needed, in sufficient quantity, and with sufficient continuity to support workflow function. A retained-demand claim rooted in operational inventory therefore points toward requirements of positioned balance, replenishment depth, and routine executable liquidity.

Collateral demand implies a different quantity problem. If an asset is to function in collateral form, the relevant issue is not only how much value exists nominally, but how much of that value can be recognized, mobilized, and posted under the applicable institutional conditions. Required balances in this case depend not simply on flow size, but on collateral eligibility, haircut treatment, mobilization speed, and the degree to which locked balances reduce immediately usable float elsewhere. A collateral-based retained-demand claim therefore connects to sizing through recognized and mobilizable balance requirements rather than through transfer flow alone.

Liquidity-buffer demand introduces yet another form of requirement. Here the asset is not held primarily for routine throughput, but for resilience against mismatch, discontinuity, or execution uncertainty. The relevant question is not how much flow the system handles in calm conditions, but how much reserve depth must exist if conditions deteriorate, timing shifts, or execution becomes less certain. This means that required liquidity conditions must be evaluated not only against average operational flow, but also against tolerated stress, buffer policy, and the size of potential mismatch under adverse conditions. A retained-demand claim rooted in liquidity buffering therefore implies required reserve depth under uncertainty rather than ordinary inventory alone.

Waiting-liquidity demand is temporally distinct again. Where workflows involve staging, pending settlement legs, conditional release, or asynchronous completion, the asset may need to be held across time even if its ultimate transfer role is temporary. In such cases, the quantity requirement depends not

only on transaction size but on temporal overlap, processing delay, concurrency of staged balances, and the extent to which capital remains immobilized before completion. A retained-demand claim rooted in waiting liquidity therefore implies required staged balances across time rather than merely high aggregate throughput.

These distinctions matter because they show that retained demand cannot be translated into quantity conditions through a single undifferentiated metric. The relevant sizing problem depends on the channel through which retained demand is said to arise. Inventory demand, collateral demand, liquidity-buffer demand, and waiting liquidity each place different pressure on balance availability, mobilization, timing, and execution depth. Any serious attempt to move from retained demand to liquidity conditions must therefore preserve these distinctions rather than flatten them into a generic notion of "more demand."

This also clarifies why the present paper does not proceed directly from retained demand to price interpretation. Between those two lies the intermediate problem of required liquidity conditions. If retained demand is said to matter, one must first identify what forms of held balance it implies, what degree of usable depth those balances require, how quickly they must be mobilized, and under what slippage tolerance they can function. Only after those questions are posed does it become possible to examine whether the surrounding liquidity environment could plausibly sustain such requirements.

For that reason, the retained-demand categories inherited from Paper 1 should be read here not as conclusions but as analytical inputs into a quantity framework. They identify the channels through which operational relevance might generate required balances. The next step is to specify the liquidity concepts needed to examine those requirements more precisely. That is the purpose of the next section, which develops gross-flow density and required liquidity density as audit concepts for relating flow size, usable float, turnover, available depth, and tolerated slippage.

## 4. Gross-Flow Density as an Audit Concept

If retained demand creates a quantity problem, the next step is to specify the terms in which that problem should be analyzed. This paper proposes gross-flow density as an audit concept for doing so. The purpose of the concept is not to redescribe price in more technical language, nor to introduce a metaphor for "large demand." It is to identify the liquidity conditions required to process a given level of flow within bounded slippage, timing, and operational constraints.

The term density is used here in an operational rather than metaphorical sense: it concerns how much usable liquidity must be available, in the relevant locations and time windows, relative to the gross flow burden placed on the asset.

The starting point is simple. A given amount of gross flow does not, by itself, tell us whether a settlement-related role is operationally supportable. The same nominal flow may be manageable in one liquidity environment and destabilizing in another. What matters is not gross flow in isolation, but the relation between flow size and the conditions under which that flow must be processed. If those conditions require positioned balances, mobilizable inventory, recognized collateral, or executable depth within bounded tolerance, then the relevant question becomes one of density rather than simple volume.

Gross-flow density, as used here, refers to the relationship between required flow processing and the usable liquidity conditions available to support it. It asks, in effect, how much executable liquidity, positioned balance, or mobilizable depth is required relative to the flow that must be absorbed under specified operating constraints. The concept is therefore relational rather than nominal. It does not ask only how much value exists in aggregate, but how densely usable liquidity must be available if a given level of operational flow is to be carried without excessive disruption.

This relational structure matters because not all nominal balances are equally usable. An asset's total notional value, total supply, or broad capitalization may be large while the amount of float actually available for settlement-relevant use remains limited. Some balances may be dormant, strategically held, encumbered, jurisdictionally trapped, operationally inaccessible, or concentrated in forms that cannot be mobilized where needed. For this reason, gross-flow density must be distinguished from nominal scale. The relevant issue is not how large the asset appears in aggregate, but how much usable liquidity is available under the actual constraints that define the workflow.

Several variables are especially important in this framework. The first is usable float: the portion of balances that can in practice be mobilized for the relevant function under the relevant institutional conditions. The second is available depth: the executable quantity that can be moved, sourced, or absorbed without breaching tolerated slippage or destabilizing market conditions. The third is turnover: the degree to which liquidity can be recycled through repeated use rather than requiring static balance growth one-for-one with gross flow. The fourth is positioning: whether balances are already located where they must be used, or whether additional transfer, staging, or mobilization is required before they become operationally effective. The fifth is slippage tolerance: the degree of execution friction that the relevant workflow can bear before the asset ceases to function as a plausible operational instrument.

Taken together, these variables clarify why liquidity density is not reducible to market capitalization. Market capitalization may influence some of these conditions indirectly, but it does not specify them. It does not tell us how much float is actually usable, how concentrated or fragmented liquidity is, how much depth is available at the relevant execution points, how rapidly balances can turn over, or how much slippage a workflow can tolerate before operational relevance breaks down. A capitalization figure may therefore be compatible with very different liquidity environments. What matters for the present framework is not nominal valuation in the abstract, but whether usable liquidity is sufficiently dense relative to required gross flow.

This is also why the concept must remain conditional. Required liquidity density cannot be defined once and for all in the abstract. It depends on the size and concentration of the relevant gross flow, the share of that flow that actually touches the asset, the degree of pre-positioning required, the extent of netting or internalization elsewhere in the system, the temporal overlap of liquidity needs, and the execution tolerances that institutions impose. The point of introducing gross-flow density is therefore not to produce a universal number, but to create a disciplined way of relating flow assumptions to liquidity requirements.

A further distinction is useful here. Gross-flow density refers to the overall liquidity conditions implied by a given level of flow. Required liquidity density refers more specifically to the density conditions that would have to be present if an asset were to support that flow in operationally meaningful form. In other words, gross-flow density frames the problem; required liquidity density specifies the implied burden placed on usable float, depth, positioning, and execution tolerance. This distinction will matter in the next

section, where the paper turns from conceptual definition to conditional sizing under slippage constraints and bounded market impact.

The analytical purpose of this concept is therefore restrictive rather than rhetorical. It prevents claims about settlement relevance from moving too quickly from structural plausibility to price language. It forces the question back onto usable float, executable depth, turnover, inventory positioning, and tolerated slippage. In that sense, gross-flow density is not a claim that a digital asset must be large in nominal terms. It is an audit concept for asking whether the liquidity conditions required by a stronger relevance claim are actually plausible under stated assumptions.

## 5. Conditional Sizing Under Slippage Constraints

The distinction introduced in the previous section -- between gross-flow density as the frame and required liquidity density as the implied burden -- now becomes operationally useful. Conditional sizing asks, in effect, under what assumptions required liquidity density remains below a threshold the asset can plausibly support, and under what assumptions it exceeds that threshold.

Once retained demand is translated into required liquidity conditions, the next question is how those conditions should be sized. This paper approaches that problem conditionally rather than categorically. It does not claim that any one level of liquidity must exist in all environments, nor that a single market outcome follows from structural relevance alone. Instead, it asks what level of usable depth, positioned inventory, and mobilizable liquidity would be required under explicit assumptions about gross flow, execution conditions, and tolerated slippage.

The central reason for this conditional approach is straightforward. Required liquidity is not an independent constant. It varies with the size of the flow that must be processed, the time window over which that flow must be executed, the share of the flow that actually relies on the asset, the degree of turnover available within the relevant workflow, the extent of pre-positioning already in place, and the level of execution friction that institutions can tolerate before the asset ceases to function as a plausible operational instrument. For that reason, sizing outputs in this paper are not treated as target prices in disguised form. They are conditional simulation outputs under stated assumptions.

More precisely, such outputs should be read as liquidity-feasibility tests rather than valuation estimates. They indicate the liquidity conditions that would have to be present for a claimed operational role to function under specified constraints; they do not indicate what market participants will, should, or must pay for the asset.

Slippage tolerance is a central variable in this framework. In practice, operational relevance is bounded not simply by whether an asset can be traded, but by whether it can be mobilized or executed within acceptable friction. If a required balance can be obtained only by incurring destabilizing price movement, excessive spread cost, or unacceptable execution uncertainty, then nominal availability does not translate into operational usefulness. The relevant threshold is therefore not abstract tradability, but executable liquidity within bounded slippage. This is why sizing in the present paper begins not from valuation claims, but from the relation between required flow and tolerated execution loss.

Bounded market impact provides the next discipline. Any significant attempt to source, deploy, or rebalance liquidity interacts with available depth rather than with notional supply in the abstract. The larger the required flow relative to executable depth, the greater the market impact implied by the operation. Conversely, the deeper and more recyclable the available liquidity, the smaller the balance burden required to support a given level of flow. What matters is therefore not only the gross amount of value associated with the asset, but the depth conditions under which balances can actually be entered, maintained, replenished, or unwound without breaching operational tolerances.

This relationship can be stated at a general level without committing the paper to a single closed-form model. For a given gross-flow assumption, required liquidity conditions rise as tolerated slippage becomes tighter, as usable float becomes smaller, as execution windows become narrower, as turnover becomes slower, and as inventory must be more heavily pre-positioned rather than recycled. They weaken as offsetting, netting, temporal smoothing, internalization, or repeated turnover reduce the amount of balance that must remain statically available at any one point in time. Conditional sizing is therefore not a claim about one fixed quantity. It is a relation among flow burden, execution tolerance, and liquidity environment.

Turnover deserves particular emphasis. A balance that can be reused repeatedly across a relevant cycle imposes a smaller static inventory requirement than a balance that must remain immobilized across long or overlapping settlement windows. The same is true of positioning. If liquidity is already located where it is needed, the required burden may be materially smaller than in a case where balances must first be transferred, staged, or sourced across fragmented environments. Concentration matters as well. A liquidity pool that appears large in aggregate may still impose severe friction if executable depth is concentrated in limited venues, limited counterparties, or limited time windows. Conditional sizing must therefore distinguish nominal scale from functionally available depth.

These distinctions also clarify why the present framework does not produce singular target numbers as its central output. The more disciplined output is a set of scenario-bound requirements. Under one set of assumptions -- higher turnover, wider tolerated slippage, lower flow concentration, and stronger internalization -- the required liquidity density may be moderate. Under another set -- tighter slippage tolerance, larger execution blocks, lower usable float, and heavier pre-positioning requirements -- it may be materially higher. The purpose of sizing, therefore, is not to collapse all possibilities into a single headline figure, but to clarify which assumptions generate which burden.

To make this structure concrete without committing to a single numerical model, consider three benchmark dimensions. First, usable float as a share of total balances: if this share is low -- because balances are dormant, encumbered, or jurisdictionally inaccessible -- then required static inventory per unit of flow rises materially. Second, effective turnover within the relevant settlement cycle: if the same balance can be recycled multiple times within the operating window, the required static position falls proportionately. Third, tolerated slippage as a fraction of execution cost: tighter tolerances imply a smaller executable block for any given depth, raising required positioning accordingly. In simplified terms, lower usable float, lower effective turnover, and tighter slippage tolerance all increase required liquidity density, while broader usable float, higher turnover, and wider tolerances reduce it. These three dimensions are not independent, but they allow scenario-bound sizing to be structured as a range of conditional outputs rather than a single fixed requirement. A scenario combining low usable float, low

turnover, and tight slippage tolerance represents the heaviest liquidity-density burden. A scenario combining broad usable float, high turnover, and wider tolerances represents the lightest. The framework does not specify which scenario is more realistic; that determination belongs to applied inquiry. What it does specify is the structure of the quantity problem.

This conditional structure is especially important in institutionally connected environments, where tolerated slippage is often narrow, liquidity needs may be time-sensitive, and operational balance requirements may be shaped by policy and workflow design rather than by speculative preference. In such settings, the relevant question is not whether a large nominal value exists somewhere in the system, but whether enough depth can be mobilized, recycled, and maintained where needed under the execution tolerances that institutions actually impose. Required liquidity density is therefore best understood as an operational threshold condition rather than as a rhetorical extrapolation from adoption claims.

For that reason, the framework developed here should be read as a disciplined bridge between structural relevance and quantity conditions. If retained demand is said to matter, the next question is how much usable liquidity that relevance would require under explicit slippage and market-impact assumptions. If that burden proves implausibly large under realistic operating constraints, then stronger relevance claims weaken. If it remains supportable under bounded assumptions, then a more serious case for operational significance can be entertained. In either case, the analytical gain comes from making the assumptions visible.

The next section turns to weakening and invalidation conditions. Its purpose is to show that stronger sizing claims do not fail only when the numbers are "too low," but whenever internalization, substitution, compression, weak retained-demand intensity, or shallow realized selection materially reduce the burden that the present section treats conditionally.

## 6. Weakening and Invalidation Conditions

A useful sizing framework must do more than generate conditional requirements under favorable assumptions. It must also specify the conditions under which stronger sizing claims weaken or fail. The present framework is therefore deliberately asymmetrical in its discipline. It does not begin by assuming that retained demand is intense, that realized selection is deep, or that external liquidity burdens necessarily remain large. Instead, it asks how much of the quantity problem survives once the relevant compressing and substituting conditions are taken seriously.

The weakening conditions specified in this section operate directly on required liquidity density. Each condition reduces the burden that gross-flow density would otherwise place on usable float, available depth, turnover, and positioning.

The first weakening condition is low retained-demand intensity. The present paper inherits from Paper 1 the distinction between visible use and held-form necessity. But even if retained-demand channels can be identified conceptually, the actual intensity of those channels may remain modest in practice. Inventory may be required only episodically rather than continuously. Collateral use may remain narrow rather than systemically broad. Liquidity-buffer demand may be small if tolerated mismatch is low or if operational environments remain stable. Waiting-liquidity burdens may be limited if staging windows are short or

sparse. In such cases, the quantity problem does not disappear categorically, but it weakens materially because the amount of balance that must remain available in held form remains limited.

The second weakening condition is shallow realized selection. Paper 2 argued that residual overflow does not by itself imply selection. The same logic applies here. Even where structural relevance remains possible, actual workflow dependence on the asset may remain thin, localized, or intermittent. If only a small share of relevant activity is actually routed through the asset, then the gross-flow burden placed on usable liquidity conditions will remain correspondingly smaller. A stronger sizing claim therefore requires not only plausible retained-demand channels in principle, but evidence that realized selection occurs at meaningful operational depth.

The third weakening condition is internalization. A quantity burden falls sharply if institutions can satisfy the relevant function within their own balance sheets, treasury systems, custody structures, or coordinated operating perimeters. If liquidity can be pre-netted, reused internally, or routed within already-integrated organizational environments, the amount of external positioned inventory required may shrink materially. The quantity problem is therefore strongest where the asset must support genuinely external or cross-boundary liquidity conditions. It weakens where the same economic burden can be absorbed inside bounded institutional architectures.

The fourth weakening condition is netting, compression, and temporal smoothing. Gross flow should not be confused with net liquidity burden. A system may process large nominal flows while requiring only a fraction of those balances in static or quasi-static form if offsetting, recycling, synchronization, or staged optimization materially reduce the amount of liquidity that must remain simultaneously available. The more effectively obligations are compressed across time, counterparties, and pathways, the lower the required density burden becomes. In this sense, high gross flow does not automatically imply high required liquidity density. What matters is how much of that flow survives compression as an actual burden on usable balances.

The fifth weakening condition is substitution by full-coverage or near-full-coverage alternatives. If stablecoins, tokenized deposits, CBDC-linked pathways, internal treasury rails, or other bounded instruments can absorb the relevant function end-to-end with sufficient admissibility and efficiency, then the external liquidity burden placed on the asset declines. This does not require perfect universality. It is enough that substitutes absorb the relevant portion of flow under the relevant operating constraints. A stronger sizing claim therefore weakens wherever external connective burden is crowded out by sufficiently effective alternatives.

The sixth weakening condition is broader usable float and higher effective turnover than initially assumed. Because the framework is conditional, burden estimates depend heavily on the fraction of balances that is operationally available, on how frequently liquidity can be recycled, and on how quickly positions can be mobilized. If actual usable float is larger, if turnover is faster, or if inventory can be reused more efficiently than expected, then the same flow burden can be supported with materially lower static requirements. A stronger sizing claim may therefore fail not because structural relevance disappears, but because the quantity burden proves smaller than initially suggested under more favorable liquidity conditions.

The seventh weakening condition is lower concentration of simultaneous need. Required liquidity density rises when demand is clustered in narrow windows, concentrated in limited venues, or dependent on large synchronized movements. It weakens when flow is temporally distributed, geographically spread, operationally staggered, or diversified across multiple channels. The relevant burden depends not only on total flow, but on overlap. If balance needs do not converge at the same points in time and space, the required density threshold may be substantially lower than a simple aggregate reading would imply.

These weakening conditions also imply corresponding invalidation pathways. A stronger sizing claim fails where retained-demand intensity remains low, where realized selection remains weak, where internalization absorbs the relevant balance burden, where compression reduces simultaneous liquidity need, where substitutes crowd out the relevant function, or where usable float and turnover are sufficiently favorable that the required density threshold no longer appears operationally demanding. In such cases, the present framework does not support a stronger inference from structural possibility to substantial quantity requirement.

This is why the paper's discipline is deliberately restrictive. The framework is not designed to produce large numbers by assumption, nor to convert any plausible relevance claim into a valuation story. Its purpose is to test what level of required liquidity condition would actually follow if stronger retained-demand claims were granted under explicit assumptions -- and to show how readily those claims weaken when the surrounding environment becomes more internalized, more compressed, more substitutable, or less operationally demanding than initially assumed.

The analytical implication is therefore two-sided. If the quantity burden remains high even after these weakening conditions are applied, the case for stronger operational relevance becomes more serious. If the burden shrinks materially once internalization, compression, substitution, shallow selection, or distributed need are taken into account, then stronger sizing claims should be correspondingly reduced or abandoned. The next section applies this discipline to an XRP-adjacent interpretation boundary, where the temptation to overread conditional sizing into direct price implication is especially strong.

## 7. XRP-Adjacent Interpretation Boundary

An XRP-adjacent context remains useful for this paper, but only under strict interpretive limits. The purpose of this section is not to derive XRP-specific target prices, nor to claim that XRP has already achieved the retained-demand intensity required for strong sizing conclusions. Its purpose is narrower: to clarify how the conditional sizing framework should be read in a context where settlement significance, bridge-like relevance, liquidity narratives, and price expectations are especially prone to being conflated.

This boundary is necessary because XRP sits close to several claims that the present paper seeks to discipline. It is often discussed in relation to cross-border settlement, bridge liquidity, institutional rails, tokenized finance, treasury infrastructure, and broader digital-asset market structure. Those associations make the case analytically useful, but they also create a high risk of overreading. A framework that merely converts those associations into implied price targets would fail the discipline established in the preceding sections.

The first interpretive limit is that conditional sizing outputs are not target prices. A scenario that indicates a high required liquidity density under certain assumptions does not by itself establish that the market will

realize, sustain, or price that condition. It only indicates what liquidity conditions would be required if the assumed retained-demand intensity, flow burden, slippage tolerance, usable float, and turnover conditions were actually present. The output is therefore assumption-bound, not predictive.

The second interpretive limit is that XRP-specific relevance cannot be imported from ecosystem adjacency. Ripple-related infrastructure, XRPL activity, stablecoin issuance, treasury tools, custody integrations, or payment connectivity may all matter as surrounding context. But none of these developments, by themselves, establishes the quantity conditions required for strong XRP-specific sizing claims. The relevant question is whether XRP itself is selected, retained, positioned, mobilized, or relied upon in ways that generate non-trivial liquidity-density requirements.

The third interpretive limit concerns the distinction between structural possibility and operational burden. Even if one accepts that some residual overflow may remain across settlement stacks, and even if one accepts that XRP could in principle occupy a connective role in some workflows, it does not follow that the resulting quantity burden is large. That burden depends on the share of relevant flow that actually touches XRP, the degree of internalization elsewhere, the amount of netting or compression available, the concentration of simultaneous need, and the actual depth and turnover of usable liquidity. Conditional sizing therefore requires explicit assumptions at every step.

This makes the XRP-adjacent context a stress environment for the framework rather than evidence in favor of a conclusion. It tests whether the framework can remain disciplined when the temptation to move from possibility to price is unusually strong. If the framework is too loose, XRP-adjacent analysis will collapse into valuation rhetoric. If it is properly constrained, it will instead separate retained-demand intensity, realized selection, required liquidity density, slippage tolerance, and weakening conditions before any stronger inference is made.

For that reason, any XRP-adjacent sizing scenario should be read as a structured sensitivity exercise. It may ask how required liquidity conditions would change under different assumptions about flow size, turnover, usable float, inventory concentration, buffer requirements, or tolerated slippage. It may show that certain assumptions generate materially higher liquidity-density requirements than others. But such scenarios remain conditional. They do not establish inevitability, nor do they convert methodological outputs into market forecasts.

This interpretation boundary also protects the series structure. Paper 1 asked whether retained demand exists as a stronger category than visible usage. Paper 2 asked whether residual overflow survives closure competition and whether any external asset can pass institutional selection filters. The present paper asks what quantity conditions would follow if retained demand became operationally meaningful. A stronger applied analysis of the Ripple stack, compression, bypass, amplification, and real-world institutional implementation belongs to a later inquiry. This paper prepares that inquiry; it does not pre-empt it.

The implication is straightforward. XRP may remain the primary motivating case in the background, but the present paper does not assert XRP-specific necessity, inevitability, or target valuation. It provides a framework for asking what liquidity-density burden would be implied under explicit and weakenable assumptions. If those assumptions fail, the sizing claim weakens. If they hold only partially, the output remains partial. If future evidence strengthens retained demand, selection, and operational burden, then

stronger applied conclusions may become possible. But they must be earned through the framework rather than imported into it.

## 8. Conclusion

This paper has argued that retained demand, if it becomes operationally meaningful, raises a quantity problem that cannot be answered by visible usage, ecosystem growth, market capitalization, or price narratives alone. The relevant question is not simply whether an asset appears structurally plausible, nor what price it should command in open markets. The relevant question is what liquidity conditions would be required if the asset were relied upon in held form for inventory, collateral, liquidity-buffer, or waiting-liquidity functions at non-trivial scale.

From that starting point, the paper made four main moves. First, it separated sizing from price narratives and argued that required liquidity conditions should be treated as a distinct analytical problem. Second, it connected the retained-demand categories developed in Paper 1 to quantity requirements in positioned inventory, mobilizable collateral, reserve depth, and staged balances across time. Third, it introduced gross-flow density and required liquidity density as audit concepts for relating gross flow, usable float, available depth, turnover, inventory positioning, and tolerated slippage. Fourth, it embedded conditional sizing within a weakenable framework that includes low retained-demand intensity, shallow realized selection, internalization, compression, substitution, favorable turnover, and distributed need as conditions under which stronger sizing claims weaken or fail.

The resulting framework is deliberately restrictive. It does not claim that structurally plausible relevance automatically produces large liquidity requirements. Nor does it claim that large conditional sizing outputs should be read as target prices. Sizing outputs in this framework are scenario-bound requirements under explicit assumptions about flow size, usable depth, turnover, positioning, and slippage tolerance. Their value lies not in producing a single headline number, but in making visible which assumptions generate which liquidity burden.

This distinction matters because institutionally connected digital assets are often discussed through compressed narratives. Usage is conflated with demand, infrastructure expansion with asset relevance, structural possibility with selection, and sizing with valuation. The series of which this paper forms a part is designed to slow those inferences down. Paper 1 separated usage from retained demand. Paper 2 separated expansion from closure and overflow from selection. The present paper separates conditional sizing from price narratives. Together, these distinctions create a more disciplined pathway from observation to claim.

The framework also clarifies how stronger sizing claims should be weakened. If retained-demand intensity remains low, the quantity burden remains limited. If selection is shallow, the flow burden placed on the asset remains small. If internalization, netting, compression, or full-coverage substitutes absorb the relevant function, the need for external positioned inventory declines. If usable float is broader, turnover is higher, or simultaneous need is less concentrated than assumed, required liquidity density may be materially lower. These are not secondary caveats. They are part of the framework's central audit discipline.

An XRP-adjacent context served here only as an interpretation boundary. It is useful because claims about settlement significance, liquidity need, and possible future valuation are especially prone to being conflated in that setting. But the present paper does not derive XRP-specific target prices, nor does it establish XRP-specific necessity. It specifies how such claims would have to be structured if they are to become analytically defensible: retained demand must be specified, selection must be real, flow assumptions must be explicit, liquidity density must be tested, and weakening conditions must be applied.

The broader contribution is methodological rather than directional. This paper provides a way to ask how much liquidity would be required if retained demand became operationally significant, while preserving the distinction between required liquidity conditions and valuation rhetoric. It therefore does not settle the fate of any one asset. It establishes a stricter framework for translating retained-demand claims into quantity conditions under explicit and weakenable assumptions.

Within the series, this paper fixes the boundary between conditional sizing and valuation. It provides the quantity layer on which later applied analyses can test compression, bypass, amplification, and real-world stack design without treating sizing outputs as price forecasts.

The next step belongs to a more applied inquiry. If Paper 1 defines the retained-demand audit category, Paper 2 analyzes the closure competition that determines whether residual relevance can survive, and Paper 3 translates retained demand into conditional liquidity requirements, then a later Paper 4 can examine how compression, bypass, amplification, and specific institutional stack design affect those requirements in practice. The present paper prepares that inquiry by clarifying the quantity layer, but it does not pre-empt the applied conclusion. Accordingly, this paper stops short of evaluating whether any specific institutional stack has in fact generated the retained-demand intensity, realized selection, and liquidity-density burden required for stronger asset-specific conclusions.

## Footnotes

1. This paper is designed to extend the analytic architecture developed in Paper 1 and Paper 2 rather than to replace it. Paper 1 separated visible usage from retained demand and decomposed retained demand into inventory, collateral, liquidity-buffer, and waiting-liquidity channels. Paper 2 examined settlement stack competition and distinguished expansion, closure, residual overflow, selection, and stronger structural relevance. The present paper asks what liquidity conditions would be required if retained demand became operationally meaningful under explicit and weakenable assumptions.

2. The term "sizing" is used here in a conditional liquidity-analysis sense. It does not refer to price prediction, investment valuation, or a target-price exercise. Sizing outputs in this paper are treated as liquidity-feasibility tests: they indicate what usable float, executable depth, turnover, positioning, and tolerated slippage would be required under stated assumptions, not what the market should or must pay for the asset.

3. Required liquidity density is introduced as an audit concept rather than as a valuation metric. It refers to the liquidity conditions that would have to be present for an asset to support a specified operational burden within bounded slippage, timing, and market-impact constraints. It is therefore distinct from market capitalization, nominal supply, or broad claims about future asset value.

4. The XRP-adjacent context is used as an illustrative stress environment because claims about settlement relevance, bridge-like utility, liquidity requirements, and possible future valuation are especially prone to being conflated in that setting. No asset-specific necessity, inevitability, or target valuation is asserted. Any stronger XRP-specific conclusion would require separate evidence of retained-demand intensity, realized selection, and operational liquidity-density burden.

## Keywords

Digital Assets; Retained Demand; Required Liquidity Density; Gross-Flow Density; Conditional Sizing; Liquidity Feasibility; Usable Float; Available Depth; Turnover; Bounded Market Impact; Slippage Tolerance; Settlement Infrastructure; Institutional Finance; Structural Audit; XRP

## Author Note

This paper is presented as a working paper and should be read as a framework-oriented contribution. It is the third paper in a retained-demand audit series for institutionally connected digital assets. Its purpose is to clarify how retained demand may be translated into required liquidity conditions when sizing, liquidity density, market impact, and price narratives are treated as analytically distinct questions. XRP appears here only as a motivating case and illustrative stress environment; the contribution of the paper is methodological rather than advocative. The paper does not provide investment advice, a price target, or a recommendation to buy, sell, or hold any digital asset.

## Short Description

A working paper proposing a conditional sizing framework for institutionally connected digital assets. The paper separates sizing from price prediction and examines how operationally meaningful retained demand may translate into required liquidity density, usable float, available depth, turnover, inventory positioning, and bounded market impact.